(a) ProCams data preparation

Gray image x_0

Project and capture

Surface image $\tilde{s}_s$

Surface

Sample

Projector input $\{x_i\}_{i=1}^{N_s}$

Captured image $\tilde{x}_{s,i}$

(b) Mobile model preparation

Distill

Teacher - - -> Student

QAT

Per-channel

INT8

Export

Geo. state

Warp

Photo. graph

QDQ

QNN

NPU

Compile

Mobile deployment optimization

(c) Target-setup initialization

MobileComp

DESKTOP GPU

SI-MobileComp

ON PHONE

Capture

$\mathcal{X}_s$

Train

λ_1

G_s

f_{θ_s}

Setup package

$\mathcal{R}_s$

$\mathcal{P}_{s,K}$

Flow

E_ϕ

Ω_s

$\mathcal{F}_s$

Setup package

λ_2

PRE-TRAINED

Deploy

(d) Per-frame mobile compensation

Desired (GT) x'_t

MediaCodec decoding

Compensation $x^*_{s,t}$

H.264

Projector

Decoded stream

Camera-captured $\tilde{x}^*_{s,t}$

Camera